

MeetHalfway AI — SIGSPATIAL 2026 Demo Track 一页定位

给导师：这是把现有项目 MeetHalfway AI 投 ACM SIGSPATIAL 2026 *Demonstration Track* 的一页定位。Demo 不要求方法上的全新理论,要求的是"一个能现场跑、把空间计算可视化出来的真实系统"——我们已经有可运行原型。**截止:**论文提交 2026-06-28(≤4 页含参考文献,标题加 [Demo] 后缀,EasyChair + ACM 模板,单盲)。**请您定夺:**① 方向是否值得投;② 是否同意把我列为一作;③ 单位/署名信息。

One-line pitch

An interactive web system that turns "where should we meet?" into a **visible spatial-reasoning pipeline**: it computes a shared *travel-time isochrone intersection* for two people, retrieves candidate venues inside that fair region, enriches them with live signals, and ranks them with an explainable, fairness-aware score.

Why it fits the Demo track

- **It is a running system, not a paper concept.** Streamlit + Folium front end, Python recommendation engine; attendees can drag two start points and watch results change live.
- **Spatial computation is the visible core.** Isochrone retrieval (OpenRouteService / Mapbox) → polygon intersection (Shapely) → reachable-overlap region → in-region candidate tagging.
- **It surfaces responsible-GeoAI tradeoffs.** Fairness penalty, privacy-separated check-in, and graceful API-failure fallbacks are all shown to the user — exactly the kind of system discussion SIGSPATIAL demos reward.

What is already implemented (grounded in current code)

Capability	In the code
Travel-time feasibility	<code>get_isochrone</code> / <code>_fetch_isochrone_ors</code> / <code>_fetch_isochrone</code> (ORS + Mapbox), circle fallback
Spatial fairness region	<code>compute_intersection</code> via Shapely <code>intersection</code> ; <code>in_isochrone_intersection</code> flag; <code>subtract_natural_barriers</code> removes rivers/highways from the overlap
Travel-time fairness gap	<code>fairness_delta_minutes = T(A,p) - T(B,p) </code> , penalized so asymmetric venues drop faster
Commute-weighted midpoint	<code>compute_weighted_midpoint</code> + <code>compute_commute_bias_weights</code> (mode/radius aware)
Multi-signal ranking	<code>final_score = distance + rating + fairness + web bonus - risk/crowd penalty + isochrone bonus + spatiotemporal bonus</code> (8 components, with <code>score_breakdown</code>)
Live venue signals	<code>async httpx</code> retrieval (Tavily / DuckDuckGo / Yelp), LLM JSON extraction with deterministic keyword fallback
POI retrieval	Mapbox → OSM Overpass → Nominatim fallback chain; ~10 venue categories
Privacy-separated flow	room-based check-in where A/B submit locations independently (<code>room_state.json</code>)
Explainability	per-venue score breakdown + natural-language reasons in the UI

Contribution claim (honest framing)

Not a new optimal-routing theorem. The contribution is the **integration**: isochrone-intersection fairness + context-aware POI retrieval + live semantic signals + explanation, combined in one robust, demonstrable workflow for two-person urban meeting decisions.

Status & what's left before 6/28

-  Working prototype,  4-page LaTeX draft already exists (`paper/sigspatial2026_demo/main.tex`),  architecture/flow figures (`docs/*.svg`).
-  2–3 polished screenshots/figures of the live UI for the paper.
-  A scripted Riverside, CA demo scenario (so it maps to the host city).
-  Fill author/affiliation, AI-use disclosure (already drafted), final proofread.

Estimated effort to submit: ~1 week of part-time work. The draft is the bottleneck-free path.